

16 | OVERVIEW



Building Community-Driven Climate Resilience in West Bengal

Abstract

In 2020, a cyclone struck West Bengal, India, causing \$13 billion in damage. For coastal villages, storms like this close schools, spread disease, ruin farmland, and harm mental health. Harvard Chan C-CHANGE and local partners are developing a community-driven approach to climate resilience that starts by listening to residents, especially young people, and then designs and tests solutions with them.

Introduction

Coastal communities in developing countries contribute very little to global carbon emissions, yet they suffer the most from climate change. When extreme weather hits West Bengal's coast, the damage reaches every part of life. Children are pulled out of school, sometimes permanently, to help their families recover. Essential documents like school certificates are lost in floods, with lifelong consequences. Critical services are interrupted, diseases spread, and farmland loses fertility. Researchers want to understand these lived experiences in order to identify solutions that are effective and long overdue, from practical fixes like digitizing school certificates to bigger policy changes.

How Do Scientists Study This?

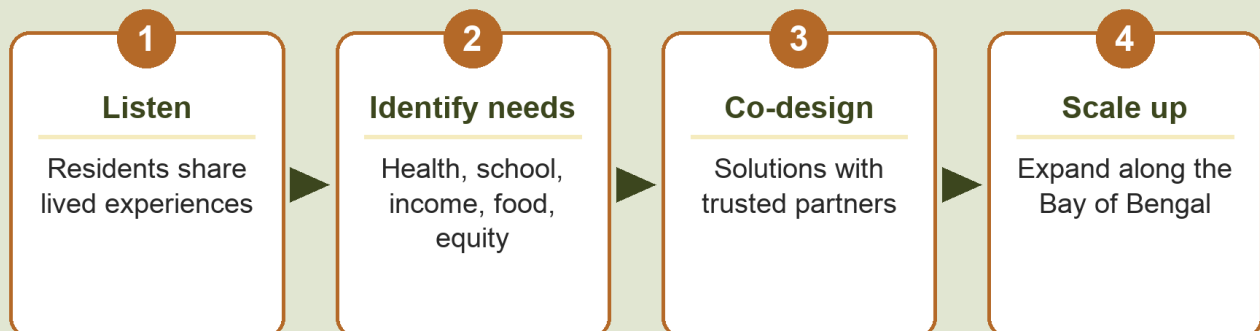


FROM THE ARTICLE

Project location: coastal West Bengal, India. Map: Harvard Chan C-CHANGE.

The project follows a community-driven design. First, researchers identify how extreme weather affects residents across six areas: health, economic disruption, displacement, education, food insecurity and malnutrition, and gender-based vulnerability. They convene community leaders, health workers, local government, and residents to discuss needs. The team is developing an interview guide and survey tool covering mental health, gender equity, education, economic stability, and environmental sustainability, to collect qualitative data about what matters most to residents. Next, they work with community members and trusted partners, including the Child in Need Institute, to implement and assess solutions. Successful programs could then be scaled along the Bay of Bengal coastline.

A community-driven approach



Study the Process

Listen

Interviews and surveys capture how storms affect health, schooling, income, food, and gender equity.

Co-design

Residents, health workers, and officials design solutions to problems they identify.

Test and scale

Solutions are assessed, then expanded along the coast if they succeed.

What Did They Discover?

The project is in its early stages, but it has already identified promising directions. One is creating jobs in ecological restoration, such as planting and preserving mangrove forests. Mangroves store carbon and protect the coast from storm surges, so this work tackles climate change, provides income, and helps communities take charge of their future. Another is practical: digitizing school certificates so they are not lost in floods. The team is paying special attention to adolescents, whose education and futures are shaped by these disruptions. By sharing its survey tools, the project hopes other communities can replicate the approach, prioritize interventions, and track progress.

Climate impacts in West Bengal

\$13B

damage from the 2020 cyclone in West Bengal

6

impact areas studied, from health to gender equity

5

survey areas: mental health, gender, school, income, environment

The researchers' top priority is understanding what matters most to the people most affected by climate change, rather than imposing outside solutions.

16 | TAKEAWAYS

Takeaways

Unequal burden

Communities that emit the least often suffer the most from climate disasters.

Disasters disrupt everything

Storms affect health, schooling, income, food, and mental health at once.

Start with people

Listening to residents leads to solutions like mangrove jobs and digital records.

Key Terms

Cyclone

A powerful rotating storm, called a hurricane or typhoon in other regions, that brings strong winds, heavy rain, and storm surges.

Mangrove

Salt-tolerant trees that grow along tropical coasts. They protect shorelines from storms and store carbon.

Carbon sequestration

The capture and long-term storage of carbon dioxide, for example by trees and soils.

Qualitative data

Information gathered through words, such as interviews and stories, rather than numbers.

Gender-based vulnerability

Ways that disasters can affect women, girls, men, and boys differently because of social roles and inequalities.

16 | ACTIVITY

Activity

Community Listening Session

Students role-play a community meeting to identify and prioritize climate resilience needs in a coastal village.

Materials: Role cards: student, parent, health worker, fisher, local official, teacher; Chart paper; Sticky dots for voting

1. Each student takes a role card and writes how a cyclone affected their character's life.
2. Share stories in a circle while a recorder sorts them into the six impact areas from the lesson.
3. Brainstorm solutions for the most common problems, including ideas like mangrove jobs or digital records.
4. Vote with sticky dots to choose the top two solutions.
5. Reflect: how might the chosen solutions differ if outsiders had decided without listening?

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Building community-driven climate resilience in West Bengal.

<https://hsph.harvard.edu/research/climate-health-c-change/research/building-community-driven-climate-resilience-in-west-bengal/>